

GIA HUY THAI

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Computer Vision | Multimodal Learning | Generative AI

EDUCATION & RESEARCH TRAINING

AI Viet Nam <i>AI Resident</i> <i>Structured training in machine learning, deep learning, computer vision, and applied AI systems</i>	Feb 2024 - Present
VNUHCM - University of Science <i>B.Sc. in Information Technology, Honors Program</i> <i>Final-year student; Expected graduation: Jul 2027</i>	Sep 2023 - Present CGPA: 3.87/4.0
Hoang Le Kha High School for the Gifted, Tay Ninh <i>Specialized in Mathematics</i>	Sep 2020 - May 2023 Grade 12 GPA: 9.5/10.0

PUBLICATION

Thai, Gia Huy et al. (June 2026). "SAGE: Shape-Adapting Gated Experts for Adaptive Histopathology Image Segmentation". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*. **Co-first author**, pp. 7337–7346. 🌐 [Project page](#).

RESEARCH EXPERIENCE

Mixture of Multimodal Experts (MoME) Research Group <i>Leader</i>	Feb 2025 - Present
• Topic 1 - Medical Image Segmentation: Improved histopathology segmentation by 4.29 Dice points over a hybrid baseline through shape-adaptive expert routing, dynamic gating, and heterogeneous backbone fusion; co-first-authored SAGE, accepted to CVPR Findings 2026, by running 12 ablation studies across 3 medical imaging datasets. • Topic 2 - LiDAR-Camera 3D Detection: Initiated a new research direction combining intra-LiDAR expert routing across point, voxel, and range representations with a cross-modal decoder for multimodal spatial reasoning. • Planned rigorous experiments and model-behavior analyses across modalities, including ablations, robustness checks, and failure-case investigation.	
VNU-HCMUS Youth Science Conference 2025 📄 <i>Member</i>	Mar 2025 - Jun 2025
• Redesigned ResNet18 downsampling with Space-to-Depth operations to improve transformation equivariance for low-resolution CIFAR-10 classification. • Achieved 80.0% accuracy while reducing model size from 11M to below 1M parameters; evaluated robustness with G-EED scores. • Analyzed the trade-off between compactness, classification accuracy, and transformation behavior to support the research report's conclusions.	
AIVN Research Group <i>Member</i>	Jun 2024 - Aug 2025
• Led 8 technical seminars and paper-review sessions on Mamba, Mixture-of-Experts, DeepSeekMoE, and efficient sequence modeling. • Synthesized 15 papers on cardiac MRI reconstruction and efficient AI systems into architecture comparisons, experiment plans, and technical presentations. • Translated new literature into implementation questions, experiment hypotheses, discussion materials, and reproducible reading notes for the group.	

SELECTED PROJECTS

Knowledge Gap Detection System <i>Backend Engineering Project</i>	May 2026 - Jul 2026
• Built FastAPI services for LLM-generated learning plans and quizzes over Neo4j knowledge graphs with PostgreSQL/SQLModel persistence. • Added quiz-based mastery updates, PII sanitization before LLM calls, and PDF/DOCX/TXT ingestion backed by MinIO object storage. • Tech stack: Python, FastAPI, Neo4j, PostgreSQL, SQLModel, Ollama/Qwen, MinIO, Docker Compose, JWT, and REST APIs.	
Visual Agentic AI with LangGraph <i>AI Viet Nam Project</i>	Mar 2026 - May 2026
• Designed a LangGraph multi-agent supervisor that routes user queries between ReAct-based Research and Vision agents through a GPT-4o-mini coordinator. • Integrated Arxiv, Wikipedia, Tavily/DuckDuckGo search, and YOLOv11 object detection for research lookup, image description, and object-counting workflows; validated the handoff state machine with multimodal inference examples. • Tech stack: Python, LangGraph, LangChain, OpenAI, GPT-4o-mini, ReAct, Arxiv, Wikipedia, Tavily/DuckDuckGo, and Ultralytics YOLOv11.	
PEFT for Multiple-Choice Question Answering <i>AI Viet Nam Project</i>	Jun 2025 - Jul 2025
• Fine-tuned a 4-bit Llama-3.2-1B model for four-choice MedMCQA by formatting questions, answer options, and ground-truth labels into instruction prompts. • Configured LoRA across attention and feed-forward projections with gradient checkpointing, 8-bit AdamW, validation, and checkpoint selection; evaluated generated answer strings through Transformers inference. • Deployed a Streamlit demo with Python, PyTorch, Unsloth, Transformers, TRL, LoRA/QLoRA, and MedMCQA.	

- Built a Vietnamese conversational assistant by instruction-tuning Llama-3.2-1B with 4-bit quantization, LoRA adapters, a 2,048-token context, and multi-turn Vietnamese chat formatting.
- Extended the pipeline from supervised fine-tuning to preference/reward modeling on chosen-versus-rejected responses and PPO-based RLHF with OpenRLHF, Ray, and checkpointed training workflows.
- Served the aligned model through a vLLM API and Gradio chat interface using Python, PyTorch, Unsloth, Hugging Face, TRL, OpenRLHF, vLLM, and Gradio.

Emoji Image Generation with Stable Diffusion

Generative AI Project

Feb 2025 - Mar 2025

- Built a three-stage data, training, and inference pipeline; used Qwen2.5-VL to replace manual captioning with detailed multimodal image-text pairs.
- Scraped and normalized emoji assets, generated descriptions of color, expression, object attributes, and background context, then investigated qualitative failure cases.
- Tech stack: Python, PyTorch, Stable Diffusion, Qwen2.5-VL, Selenium, Hugging Face Diffusers, U-Net denoising, and noise scheduling.

Text-Guided Image Generation with Conditional Flow Matching

Generative Modeling Project

Dec 2024 - Jan 2025

- Implemented a diffusion-alternative U-Net that predicts velocity fields from noisy-clean interpolations and conditions generation on 768-dimensional SentenceTransformer embeddings.
- Trained the pipeline for text-to-image generation and extended it to instruction-guided image editing using source images and TEDBench-style edit prompts, sampling trajectories with Euler ODE integration over 100 time steps.
- Tech stack: Python, PyTorch, torchcfm, SentenceTransformers, U-Net, Euler ODE integration, and Hugging Face Datasets.

VectorRAG for Vietnamese Medical-Legal QA

Text Mining Project

Oct 2024

- Built an end-to-end Vietnamese RAG pipeline on ViHERMES with dense retrieval, BM25 hybrid search, cross-encoder reranking, and Qwen2.5-7B-Instruct generation.
- Achieved BERTScore 0.8179 and Recall@5 0.9183 by iterating on retrieval composition, reranking, prompt design, and answer evaluation through error analysis.
- Tech stack: Python, ViHERMES, bge-m3, BM25, bge-reranker-v2-m3, Qwen2.5-7B-Instruct, Ollama, and prompt engineering.

CLIP-Based Video Retrieval for Known Item Search

AI Challenge Hackathon

Sep 2024

- Indexed video-frame representations with CLIP ViT-B/16, ViT-B/32, and ViT-L/14 backbones to support natural-language known-item search.
- Built a text-to-video retrieval workflow that matched natural-language queries against indexed frame representations to retrieve relevant content.
- Tech stack: Python, CLIP, ViT-B/16, ViT-B/32, ViT-L/14, visual embeddings, and text-to-video retrieval.

Image Depth Estimation

AI Viet Nam Project

Aug 2024

- Generated stereo disparity maps with 18% lower matching error using pixel-wise and window-based correspondence search.
- Benchmarked $\mathcal{L}_1$, $\mathcal{L}_2$, and cosine-similarity costs and analyzed failures under texture and lighting shifts.
- Tech stack: Python, OpenCV, NumPy, stereo vision, correspondence search, and image reconstruction evaluation.

RESEARCH SKILLS

Research Interests	Visual perception and understanding, multimodal learning, generative and creative AI, agentic AI, trustworthy AI, mixture-of-experts
Methods	Vision Transformers, VLMs, CLIP retrieval, image generation and editing, diffusion models, conditional flow matching, segmentation, expert routing, model evaluation
Frameworks	Python, PyTorch, Hugging Face Transformers/Diffusers, OpenCV, SentenceTransformers, scikit-learn, NumPy, Pandas
Evaluation	Ablation studies, robustness evaluation, failure-case analysis, reproducibility, technical writing
English	IELTS 6.5 (Jun 2026)

SELECTED AWARDS

Smart Software Innovation Program 2026 (SSIP 2026) Outstanding Project	Jul 2026
Certificate of Merit recognizing the Knowledge Gap Detection project as an Outstanding Project.	
Data for Life Season 3 Special Award	Dec 2025
Special Award for a real-time flood forecasting system integrating NASA rainfall data and local DEMs.	
Mobius Ribbon Challenge First Prize	Jul 2025
First Prize in a faculty-level Mathematics and Computer Science competition.	
Faculty of Information Technology, HCMUS Dean's List 2023-2024	Dec 2024
Dean's List recognition for academic performance in 2023-2024.	
Rencontres du Vietnam Vallet Fellowship 2023	Aug 2023
Vallet Fellowship for academic excellence and research potential.	
Vietnam National Olympiad Third Prize in Mathematics	Mar 2023
Third Prize in Mathematics.	